

ASCENSION, Saint Helena

WMO: 619020

Lat: 7.9703S

Lon: 14.4055W

Elev: 259

StdP: 14.56

Time Zone: 0.00 (E00)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	70.0	70.9	58.1	72.9	76.3	59.2	75.7	76.2	25.7	75.1	23.4	75.6	14.6	120	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.4	86.4	75.2	86.0	75.0	85.2	74.6	77.2	83.3	76.5	82.8	76.0	82.4	18.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	134.1	80.5	74.6	130.9	80.5	73.8	127.0	80.6	40.6	83.6	40.1	82.8	39.6	82.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	22.5	21.2	68.0	88.2	1.8	0.9	66.7	88.9	65.6	89.4	64.6	90.0	63.2	90.7		
			61.6	79.3	1.3	1.4	60.6	80.3	59.9	81.1	59.2	81.9	58.3	82.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	78.9	80.6	81.9	81.9	80.8	78.9	77.3	75.7	75.2	75.5	76.1	77.3
	DBStd	2.65	1.06	1.05	1.13	1.05	1.09	1.12	1.22	1.35	1.20	1.15	1.11	1.03
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10338	895	858	989	956	954	868	845	798	755	791	782	847
	CDD65	4863	430	438	524	506	489	418	380	333	305	326	332	382
	CDH74	32788	2999	3890	5256	5097	4467	3013	1940	1128	846	1004	1198	1950
	CDH80	4864	373	754	1256	1146	765	288	88	18	8	16	33	119
(14) Wind	WSAvg	15.7	14.9	14.7	15.1	16.1	16.0	16.3	16.5	16.2	15.6	15.5	15.9	15.5
(15) Precipitation	PrecAvg	4.6	0.2	0.2	0.6	0.9	0.3	0.5	0.4	0.4	0.4	0.3	0.2	0.2
	PrecMax	9.5	0.7	0.8	2.1	3.8	1.1	1.3	1.3	1.1	1.2	1.3	0.9	0.7
	PrecMin	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	0.2	0.2	0.6	1.1	0.3	0.4	0.3	0.3	0.2	0.3	0.2	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.8	86.4	88.0	87.5	86.2	84.4	82.7	81.2	80.9	81.0	82.2	82.8
		MCWB	73.5	75.1	75.9	75.7	74.4	71.9	70.2	69.2	69.1	69.3	70.1	71.5
	2%	DB	83.9	85.8	86.6	86.3	85.3	83.2	81.9	80.6	79.9	80.4	80.9	82.3
		MCWB	73.1	74.8	75.4	75.3	73.9	71.7	70.1	69.1	68.8	69.1	69.8	71.4
	5%	DB	82.8	84.6	86.1	85.8	84.4	82.6	80.9	79.6	79.0	79.2	80.2	81.2
		MCWB	72.6	74.2	75.2	75.1	73.5	71.5	69.8	68.9	68.6	68.9	69.7	71.0
	10%	DB	82.3	84.1	85.3	84.8	83.9	82.0	80.3	78.8	78.3	78.6	79.0	80.6
		MCWB	72.4	74.0	75.0	74.9	73.3	71.3	69.6	68.7	68.5	68.8	69.6	70.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.8	77.4	78.1	78.1	76.7	75.1	73.3	72.4	71.9	72.0	72.7	74.0
		MCDB	81.2	83.3	84.5	83.9	82.4	81.2	78.6	77.2	77.3	77.9	77.9	79.8
	2%	WB	74.7	76.3	77.2	77.2	75.8	73.9	72.2	71.3	71.0	71.1	71.9	73.0
		MCDB	80.7	82.3	83.7	83.1	82.0	80.2	78.2	76.8	76.4	76.8	77.5	78.9
	5%	WB	74.1	75.5	76.5	76.5	75.2	73.2	71.6	70.6	70.2	70.6	71.2	72.4
		MCDB	80.3	81.9	83.1	82.7	81.7	79.8	78.0	76.3	75.6	76.2	76.9	78.5
	10%	WB	73.5	74.9	75.9	75.9	74.6	72.6	70.9	70.0	69.8	70.0	70.7	71.8
		MCDB	79.8	81.5	82.6	82.4	81.4	79.4	77.6	75.9	75.2	75.7	76.4	78.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.2	8.0	7.4	7.0	6.8	7.0	7.1	7.4	7.5	7.6	7.6
		MCDBR	8.4	8.6	8.4	7.8	7.4	7.2	7.4	7.6	7.9	8.0	8.3	8.2
		MCWBR	3.2	3.2	3.2	3.0	3.2	3.5	3.7	3.6	3.8	3.7	3.7	3.5
	5% WB	MCDBR	8.1	8.4	8.0	7.4	7.0	6.7	7.0	7.2	7.4	7.7	7.6	7.7
		MCWBR	3.5	3.4	3.4	3.3	3.3	3.7	3.9	3.8	4.0	3.9	3.6	3.5
(40) Clear-Sky Solar Irradiance	taub	0.419	0.423	0.414	0.404	0.392	0.391	0.409	0.460	0.493	0.442	0.396	0.412	(40)
	taud	2.435	2.435	2.495	2.526	2.521	2.490	2.427	2.249	2.127	2.313	2.495	2.448	(41)
	Ebn at Noon	292	290	288	283	277	272	269	262	262	281	297	293	(42)
	Edh at Noon	39	39	36	33	32	32	35	43	51	43	36	38	(43)
(44) All-Sky Solar Radiation	RadAvg	2177	2244	2191	1996	1816	1694	1675	1676	1745	1784	1846	2005	(44)
	RadStd	53	57	54	62	46	48	40	78	90	66	79	69	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page